Evaluation of the quality of deposited datasets in the Metabolomics Workbench data repository

Robert M Flight*, P Travis Thompson and Hunter NB Moseley

2026-10-16

Acknowledgements


Hunter Moseley

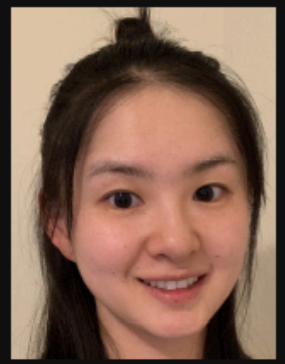

Robert Flight

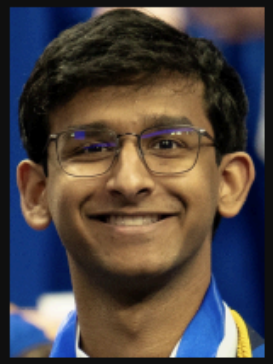

Travis Thompson


Erik Huckvale

Former Lab Members


Christian Powell


Huan Jin


Praneeth Bhatt


Joshua Mitchell


Eugene Hinderer


Andrey Smelter

Moseley Group, current and past members.

This work was supported by: NSF 2020026 (PI Moseley); NIH P42 ES007380 (PD Pennell); NIH 1R03LM014928-01 (PI Moseley)

Results of This Work

- Hosted on Zenodo at 10.5281/zenodo.23269018.
- Includes:
  - A single table with much of the summary data for each dataset, as well as the statuses and metrics of our quality control and quality assessment (QC/QA) criteria.
  - HTML QC/QA reports for every dataset.
  - Processed summarized experiments saved as R serialized rds files for every dataset.

Omics Data Repositories

- Journals and grant agencies require processed data to be deposited in a repository.
- Gene expression:
  - Gene Expression Omnibus (GEO)
  - Sequence Read Archive (SRA)
  - ArrayExpress
- Proteomics:
  - Proteomics Identifications (PRIDE)
  - Mass Spectrometry Interactive Virtual Environment (MassIVE)
- Metabolomics:
  - MetaboLights
  - Metabolomics Workbench

Metabolomics Workbench (MW)

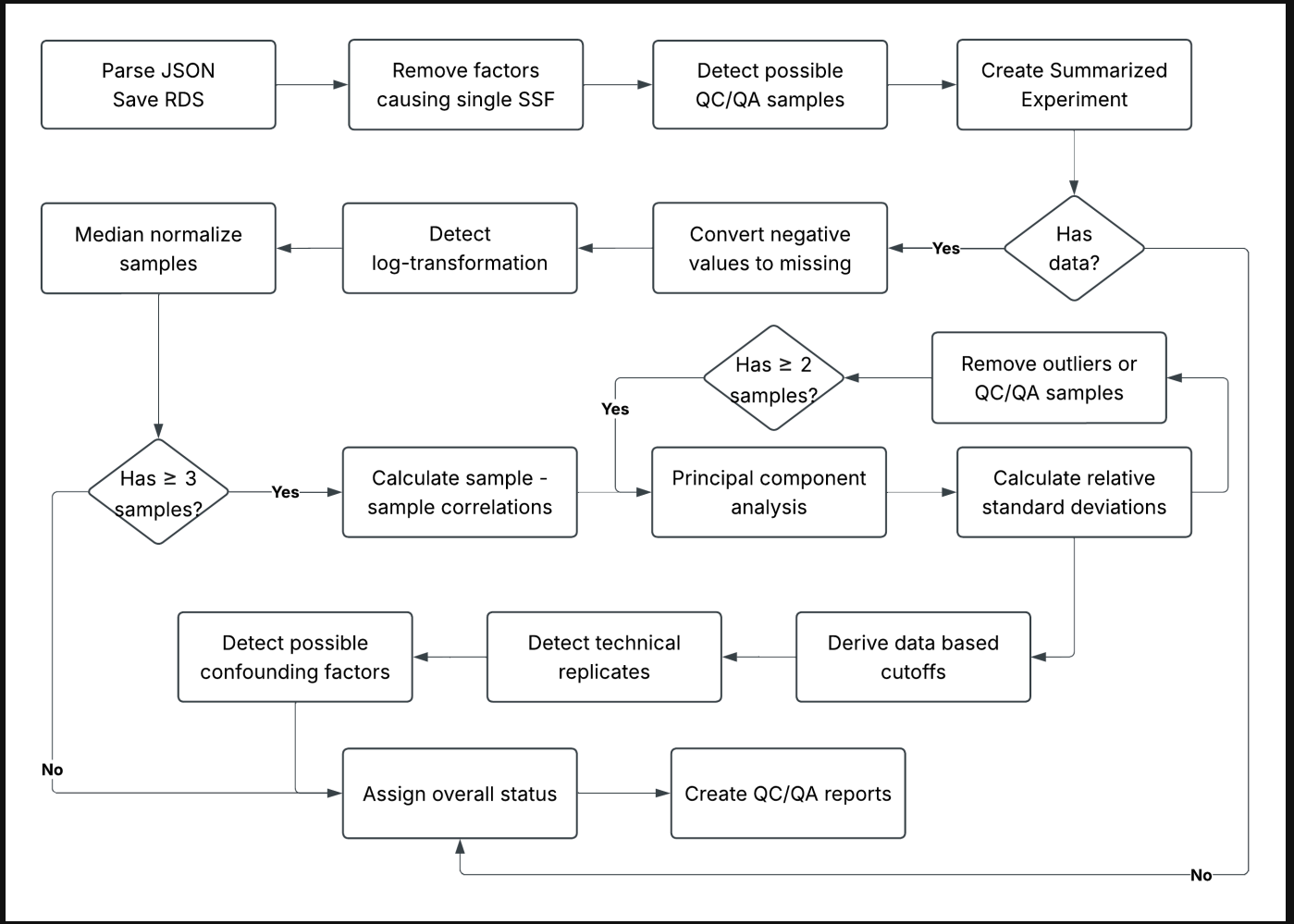
- NIH Common Fund's National Metabolomics Data Repository.
- Established in 2012.
- Currently hosts 4261 studies, from 60 countries.
- Organized by:
  - Project: a biological question, contains one or more **related** studies.
  - Study: contains one or more analyses.
  - Analysis: typically one analysis method or sample type, e.g. NMR, MS + chromatography, +ve / -ve mode, etc.
- Our lab has years of experience working with MW data.

Hypothesis: There Are Significant Numbers of Reusable Datasets in Metabolomics Workbench

- The majority of data within MW is expected to be of good quality.
- Many of the datasets within MW are part of peer-reviewed, published studies.
- Researchers depositing data into MW are trying to do good science.

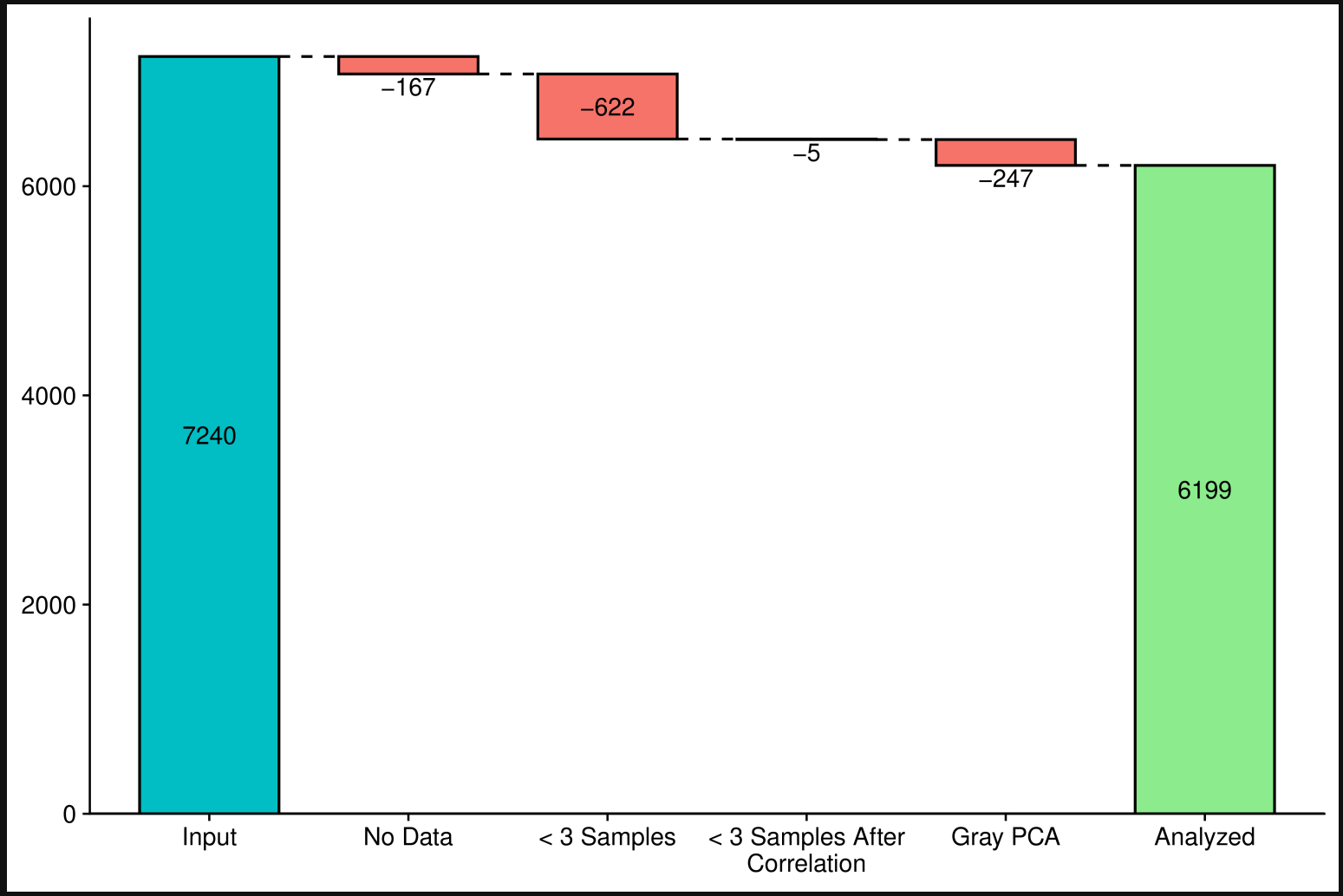
QC/QA Pipeline

- 7240 analysis datasets (2026-07-21).
- 7000 lines of R.
- 230 custom functions.
- Execution workflow managed by the `targets` R package.
- Took 5 days of execution.
  - Single machine, no HPC.



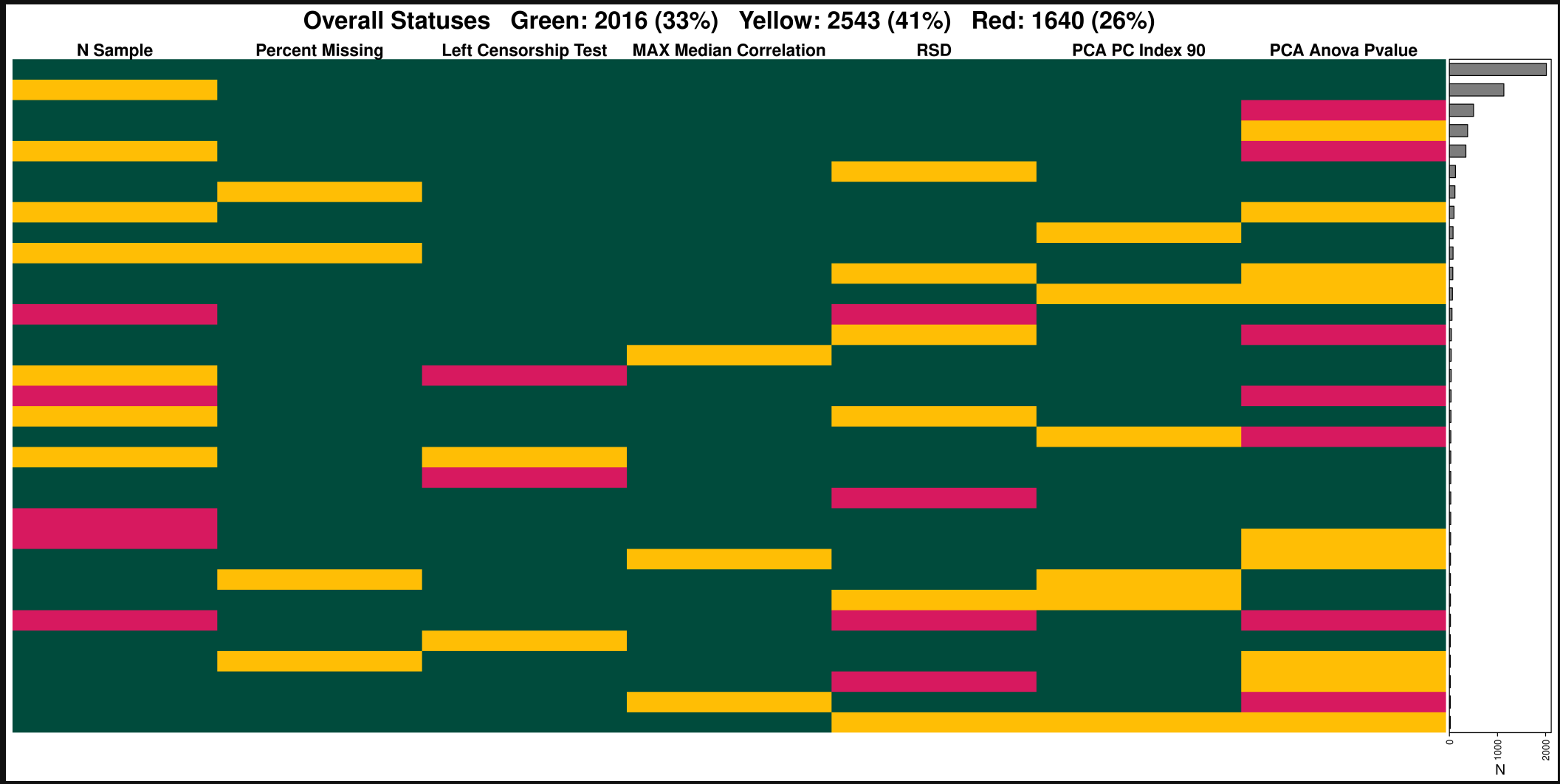
Overall QC/QA pipeline steps.

Reasons Datasets Were Removed



Color Grade Evaluation

Overall Color Status



Top 90% of status combinations.

Subject Sample Factors (SSFs)

- One or more experimentally important metadata about the samples.
  - E.g. Cancerous vs normal, high-fat diet vs normal diet, etc.
- We combine multiple SSFs to make a unique label that defines the **experimental groups** of samples.
- Additional sample data includes things like the raw file name, biological replicate.

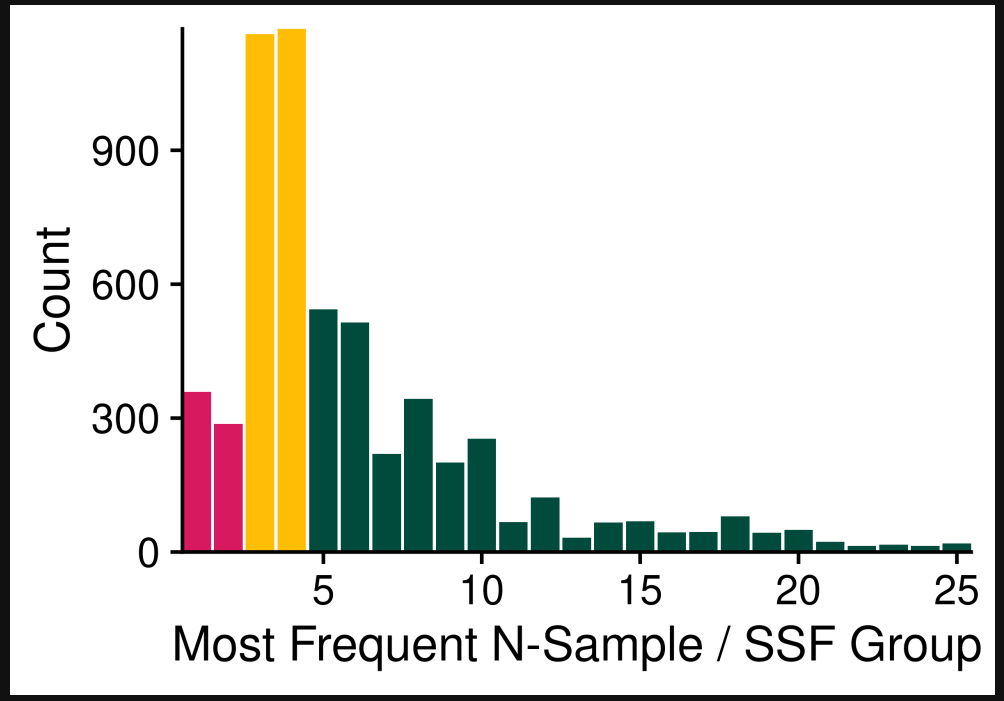
sample_id	raw_file_name_raw_file_name	cohort	group	sample_source	ssf
e09u	B_01_B_E09U_11_Saliva.mzML	First Trimester	**GDM**	**Saliva**	**GDM|Saliva**
e0jr	B_05_B_E0JR_389_Saliva.mzML	First Trimester	**GDM**	**Saliva**	**GDM|Saliva**
e0jt	B_03_B_E0JT_195_Saliva.mzML	First Trimester	**GDM**	**Saliva**	**GDM|Saliva**
e0jv	B_02_B_E0JV_100_Saliva.mzML	First Trimester	**GDM**	**Saliva**	**GDM|Saliva**

Sample metadata from AN006863.

Number of Samples in SSF Groups

- The number of samples per SSF group indicates if the experimental design is statistically powerful or not, based on general rule of thumb.
- The counts are **after** removing samples with < 10 present metabolites and any outliers.

$$\text{status}_n = \begin{cases} \text{red} & \text{if } N \leq 2 \\ \text{yellow} & \text{if } N > 2 \ \& \\ \text{yellow} & \text{if } N \leq 4 \\ \text{green} & \text{if } N \geq 5 \end{cases}$$



Using Data Distributions to Define Color Grades

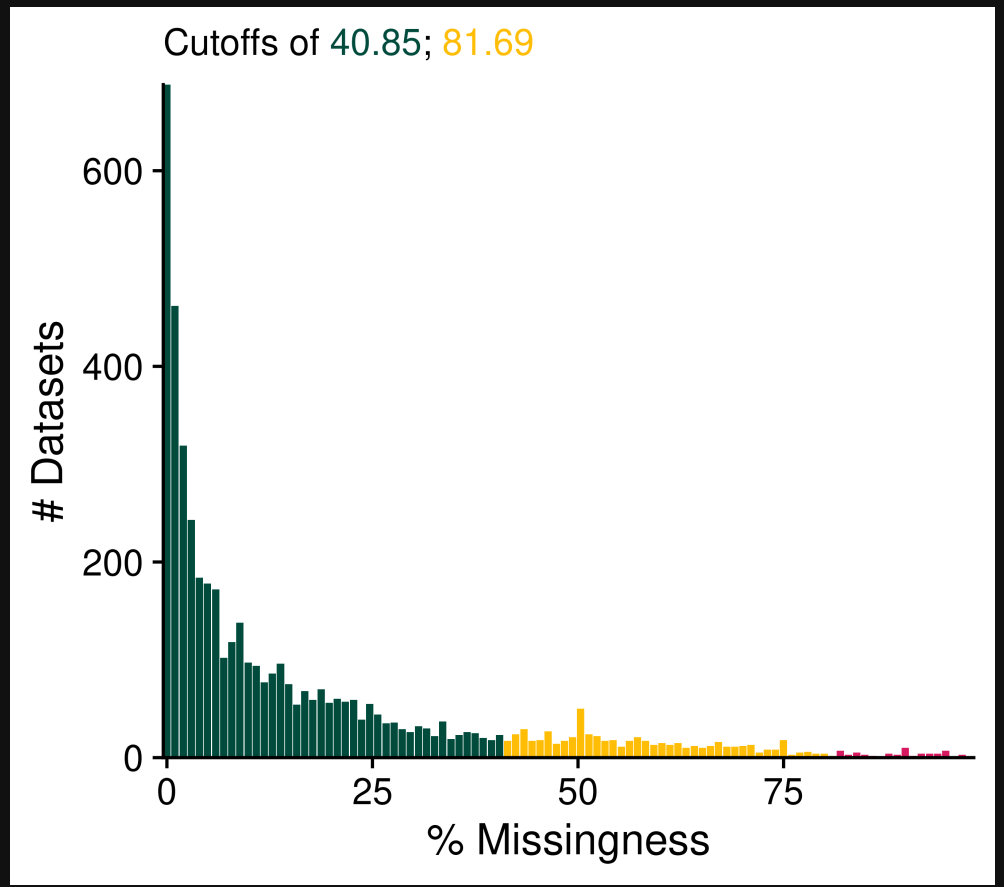
- We can use the full set of datasets themselves to define green (good), yellow (useable), red (double check).
- $\tilde{x}$, median of values.
- SD, standard deviation of values.

Percent Missing and Left-Censorship (LC)

High percentage of missing values, especially without the indication of left-censorship (LC); indicates the presence of possible data quality issues.

$$\text{status}_{pm} = \begin{cases} \text{green} & \text{if } x \leq \tilde{x} + 2SD \\ \text{yellow} & \text{if } x > \tilde{x} + 2SD \ \& \\ \text{yellow} & \text{if } x \leq \tilde{x} + 4SD \\ \text{red} & \text{if } x > \tilde{x} + 4SD \end{cases}$$

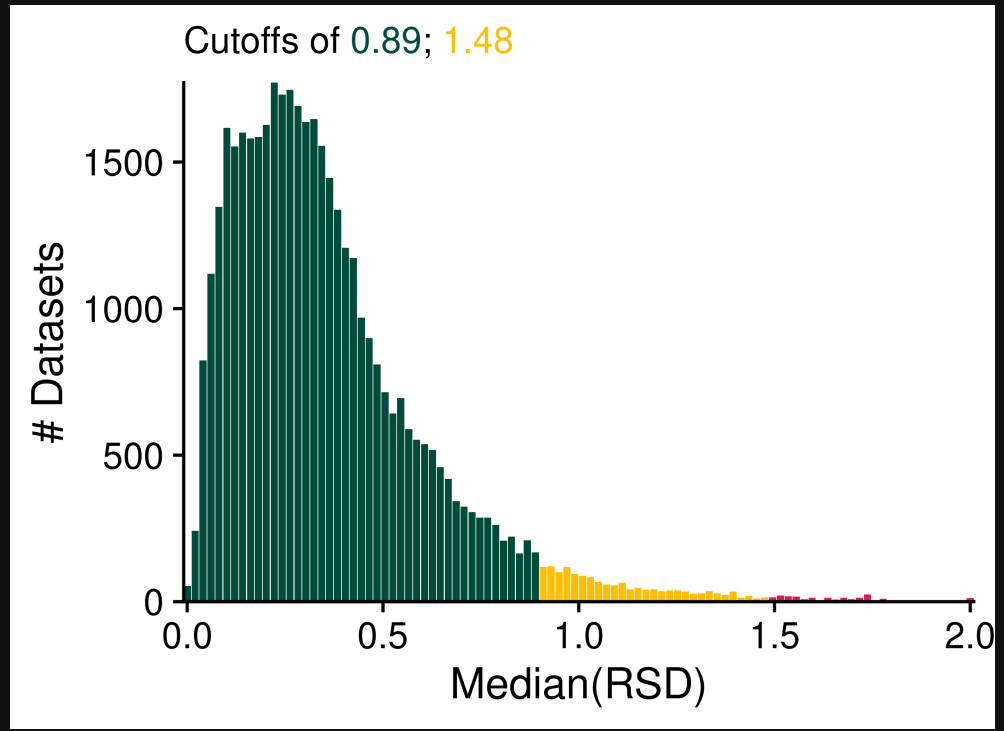
$$\text{status}_{lc} = \begin{cases} \text{green} & \text{if pm} > 10\% \ \& \ \text{P}_{LC} \leq 0.05 \\ \text{yellow} & \text{if pm} > 10\% \ \& \ \text{P}_{LC} > 0.05 \end{cases}$$



Maximum Observed Median Relative Standard Deviation (RSD)

High maximum observed median RSD indicates high variability in the measurements.

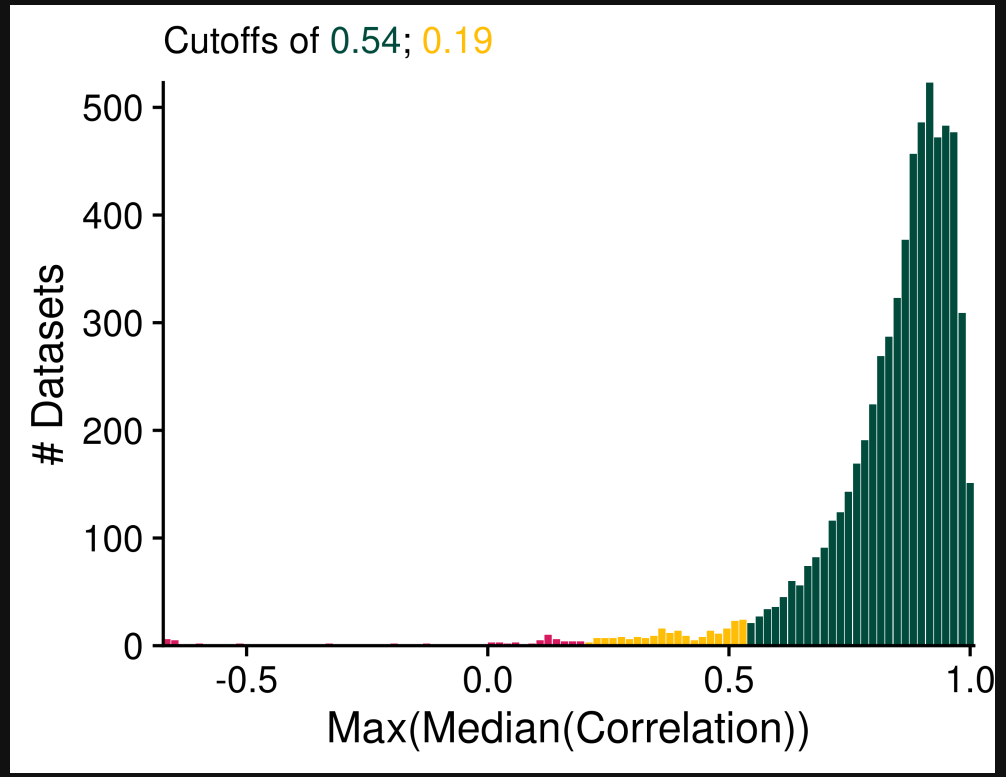
$$\text{status}_{rsd} = \begin{cases} \text{green} & \text{if } x \leq \tilde{x} + 2SD \\ \text{yellow} & \text{if } x > \tilde{x} + 2SD \ \& \\ \text{yellow} & \text{if } x \leq \tilde{x} + 4SD \\ \text{red} & \text{if } x > \tilde{x} + 4SD \end{cases}$$



Median Within SSF Group Sample-Sample Correlations

Low maximum median within SSF group sample-sample ICI-Kt correlations indicate high biological variance between samples.

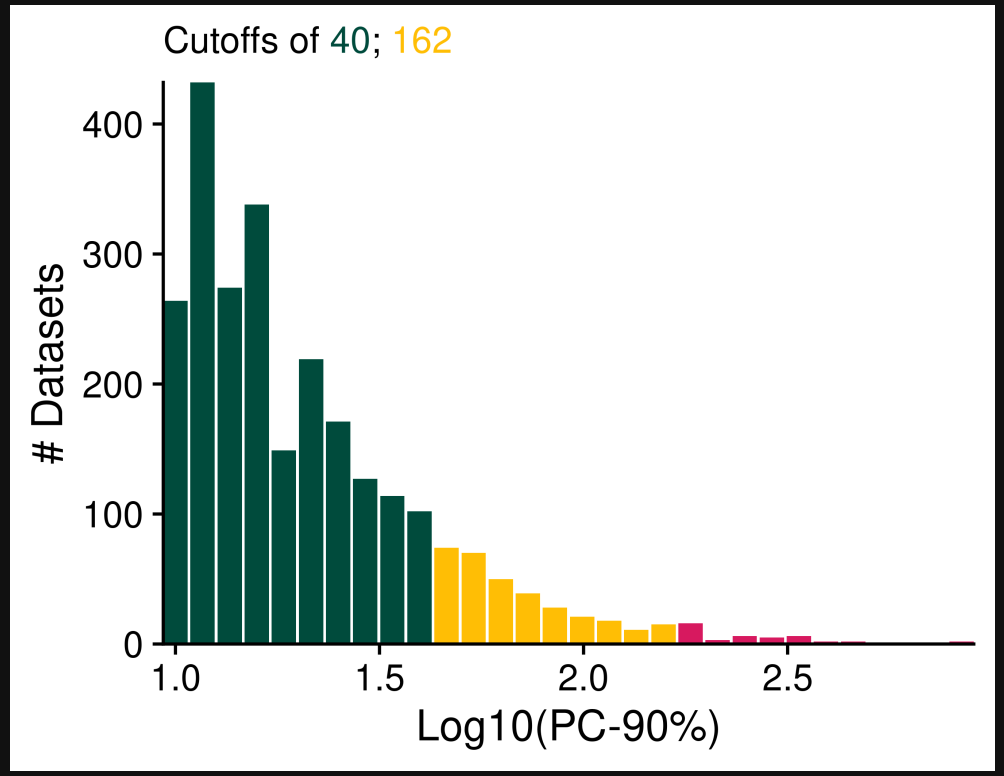
$$\text{status}_{cor} = \begin{cases} \text{green} & \text{if } x \geq \tilde{x} - 2SD \\ \text{yellow} & \text{if } x < \tilde{x} - 2SD \ \& \\ \text{yellow} & \text{if } x \geq \tilde{x} - 4SD \\ \text{red} & \text{if } x < \tilde{x} - 4SD \end{cases}$$



Principal Component Analysis (PCA) Scores - PC-90

As more principal components (PC) are needed to account for 90% of the variance in the data (PC-90), it indicates weak signal variance within the dataset in comparison to other sources of variance present.

$$\text{status}_{90} = \begin{cases} \text{green} & \text{if } x \leq \tilde{x} + 2SD \\ \text{yellow} & \text{if } x > \tilde{x} + 2SD \ \& \\ \text{yellow} & \text{if } x \leq \tilde{x} + 4SD \\ \text{red} & \text{if } x > \tilde{x} + 4SD \end{cases}$$



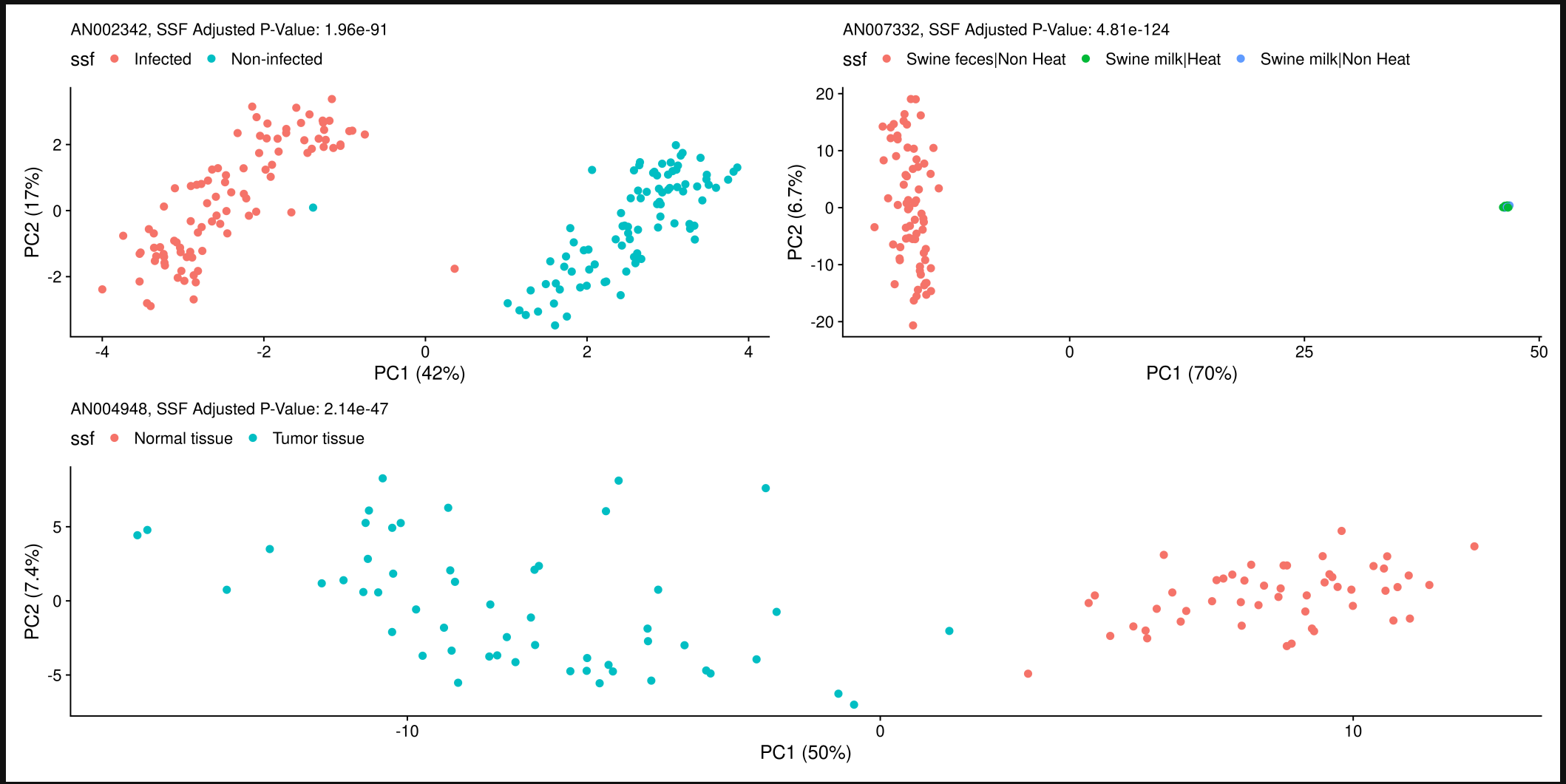
PCA Scores - ANOVA

Lack of separation of SSF groups across meaningful PCs indicates weak signal variance within the dataset.

Analysis of Variance (ANOVA) p-adjusted:

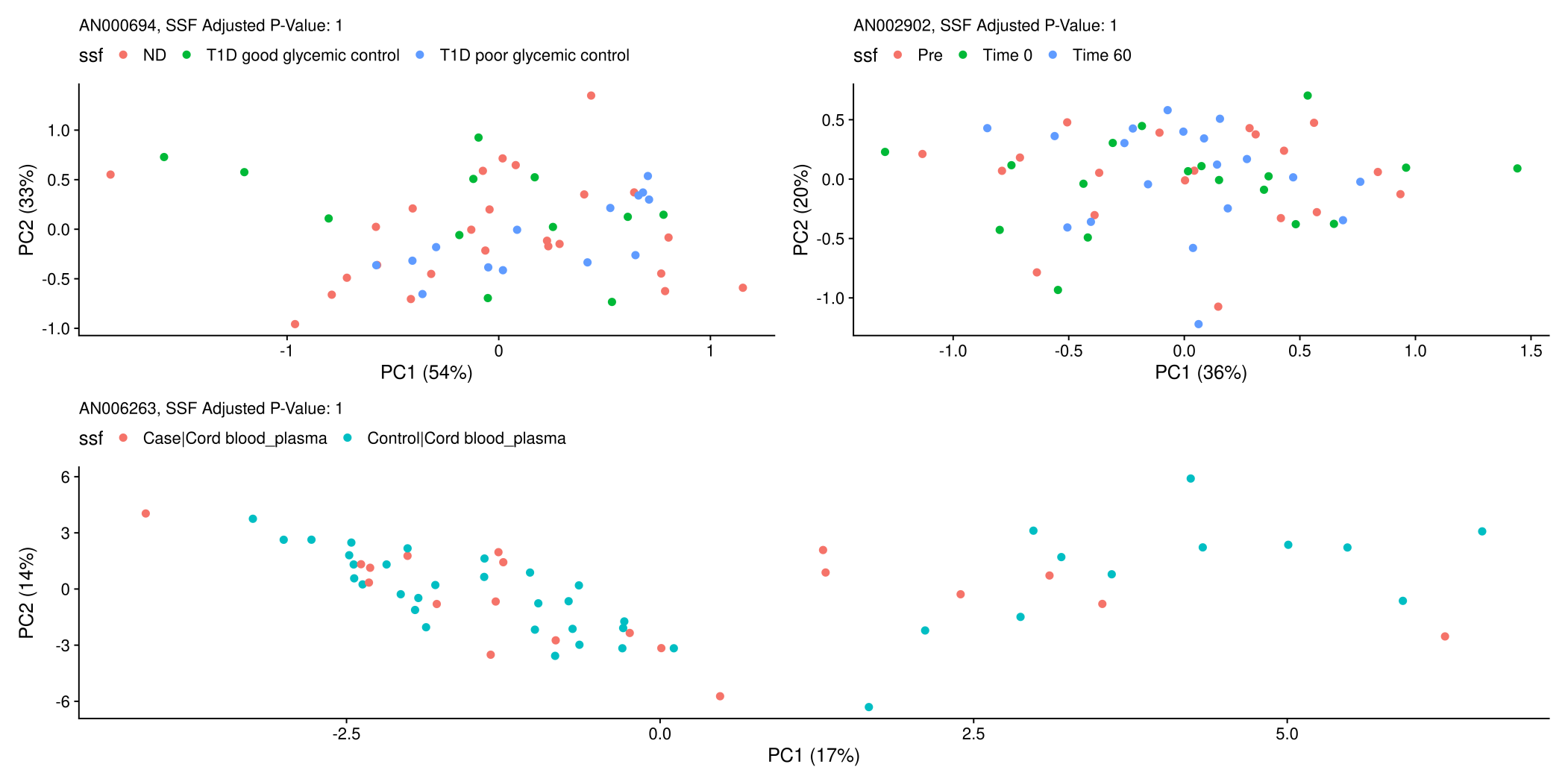
$$\text{status}_{anova} = \begin{cases} \text{green} & \text{if SSF } p_{adj} \leq 0.05 \ \& \ \text{PCperc} > 5\% \\ \text{yellow} & \text{if SSF } p_{adj} \leq 0.05 \ \& \ \text{PCperc} > 1\% \text{ or} \\ \text{yellow} & \text{if Add } p_{adj} \leq 0.05 \ \& \ \text{PCperc} > 5\% \\ \text{red} & \text{if SSF } p_{adj} \leq 0.05 \ \& \ \text{PCperc} < 1\% \\ \text{red} & \text{if SSF } p_{adj} > 0.05 \end{cases}$$

Green: PCA ANOVA P-Value



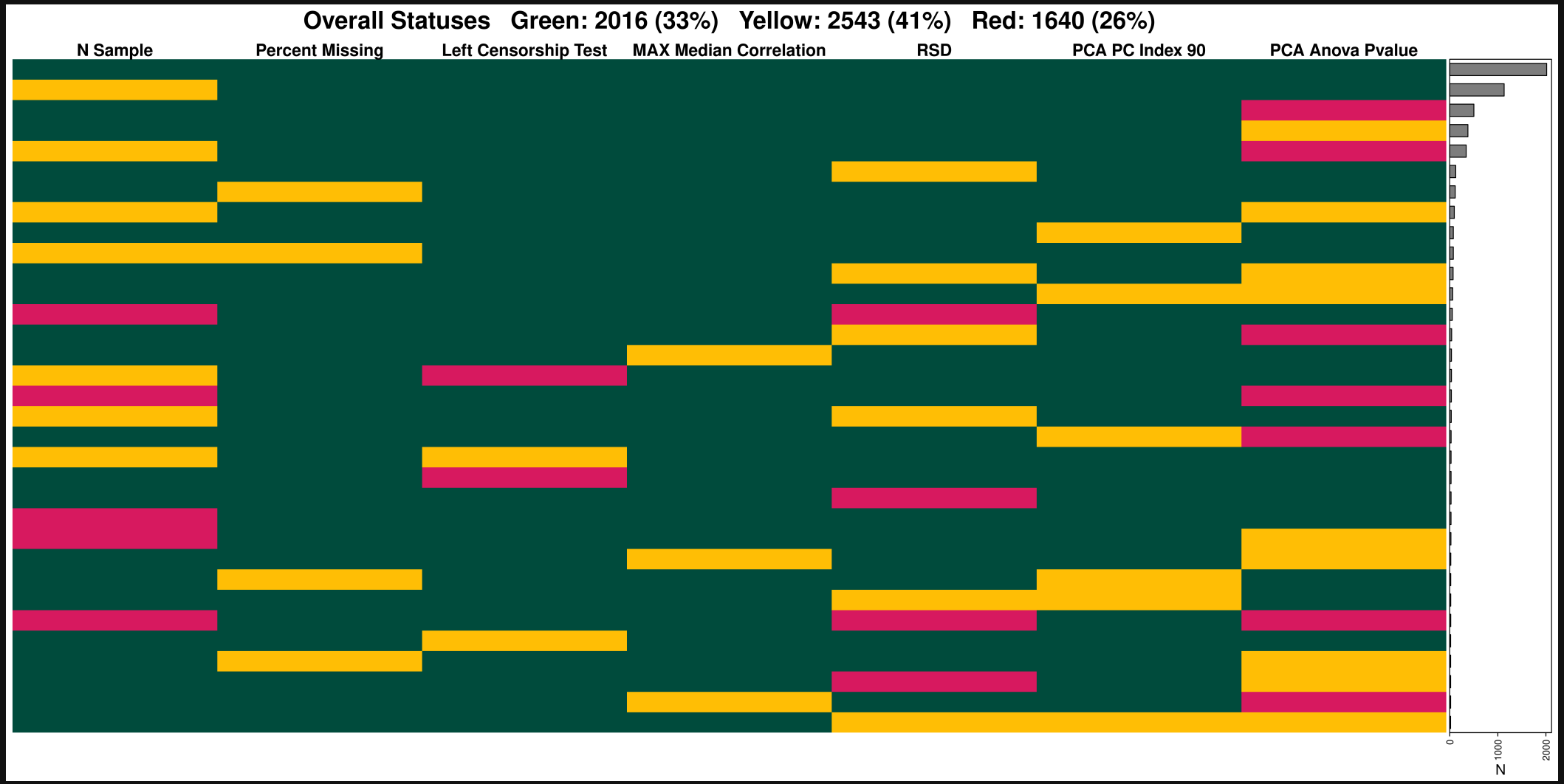
Examples of low p-value PC scores vs SSF ANOVA (green).

Red: PCA ANOVA P-Value



Examples of high p-value PC scores vs SSF ANOVA (red).

Overall Color Status



Top 90% of status combinations.

Do Analyses Within The Same Study Share the Same Overall Status?

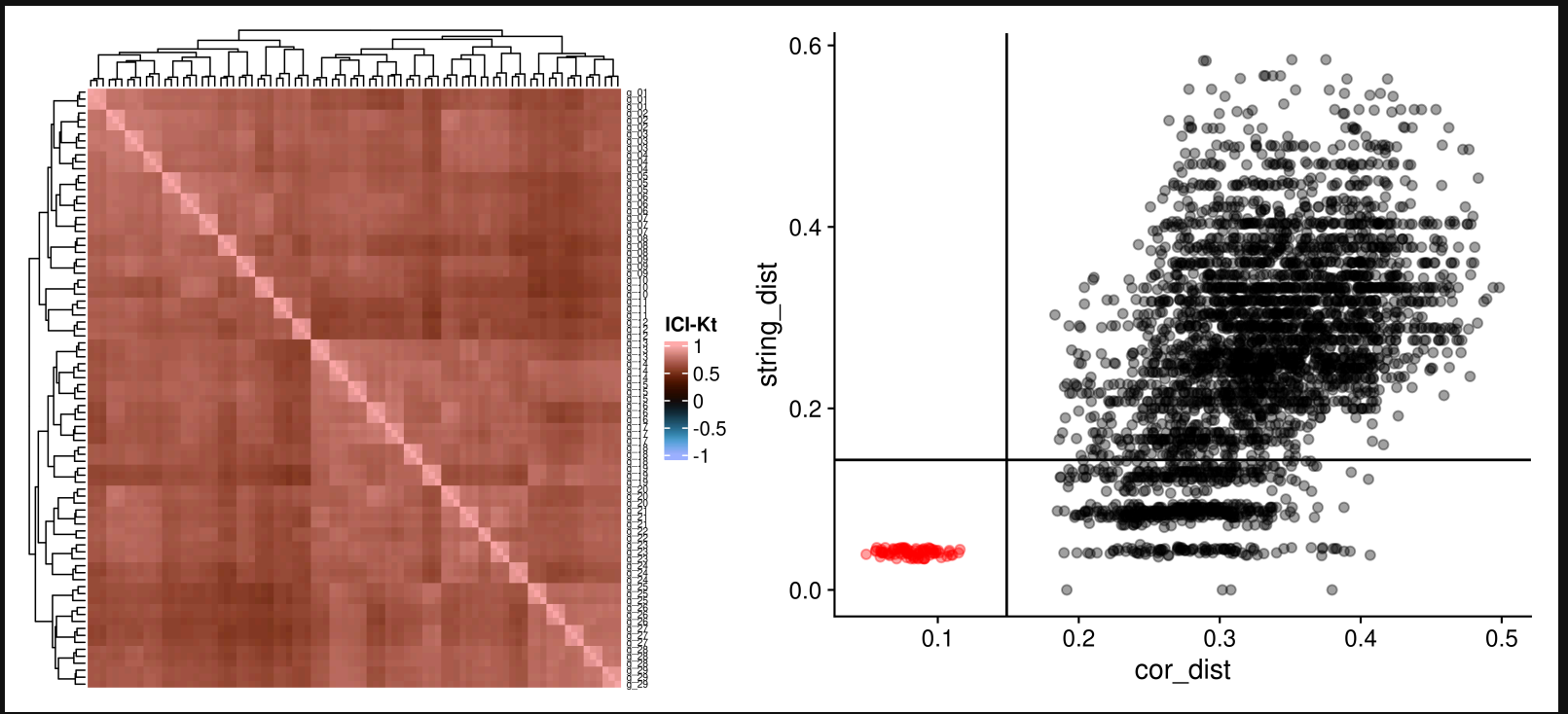
- MW groups analyses into studies;
  - multiple polarities for MS
  - multiple types of samples (blood, plasma, saliva)
  - etc

Initial status of dataset vs compared status with another dataset in the same study.

initial_status	green	yellow	red
green	813	209	111
yellow	174	1177	205
red	90	201	630

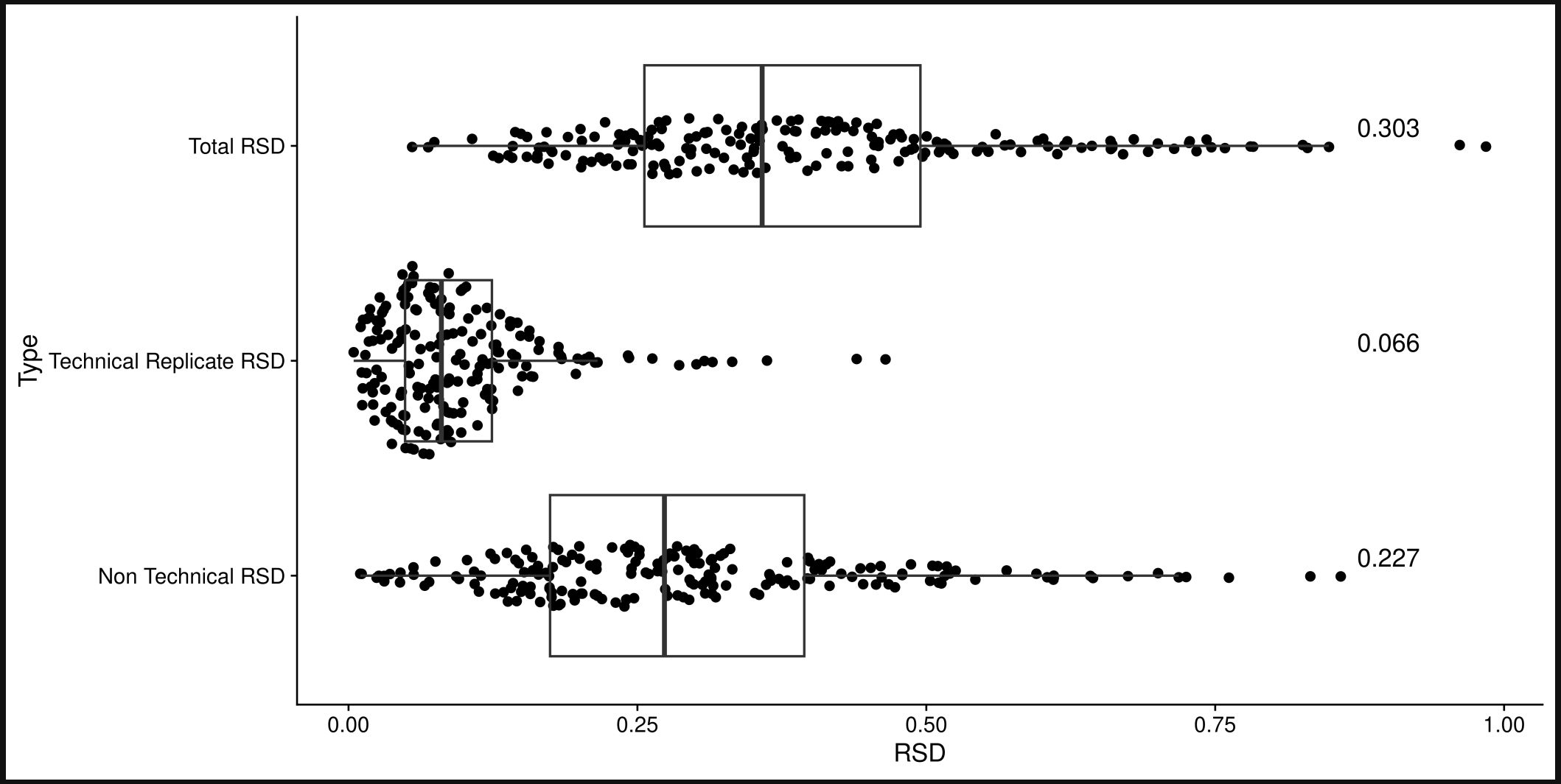
Technical Replicates Detection

- Low correlation distance
- Low string distance
- Runs in clustering by correlation distance



Sample-sample correlation heatmap, and point plot of correlation distances and string distances between samples, with replicates noted in red.

Allows Estimations of Different RSD



Sina plot of total RSD, technical RSD, and non-technical RSD for datasets with technical replicates.

Other Considerations

- Why are there datasets with no data?
- Detecting log-transformed data.
- What should be done with negative values?
- Detecting possible confounds or batch effects.

Thank you!